

Flow Matching for Thermal-Driven Wood Design

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Abstract

In computational materials science, inverse material design seeks to discover physical structures that yield specific desired properties or behaviors. This study tackles thermal-driven wood design by mapping two-dimensional macroscopic thermal responses directly back to high-fidelity microscopic wood structures. To preserve exact physical boundaries without the blurring effects of latent-space compression, we leverage pixel-space architectures (e.g., Pixel Diffusion Transformers) conditionally guided by DINOv3 semantic features. Project: <https://zekifayes.github.io/fmwd>

1 Introduction

In computational materials science, the ultimate objective of materials engineering is the reverse: discovering a structure that yields expected physical properties and behaviors [1, 2], i.e., inverse material design. In the context of bio-inspired materials like wood, current research has explored propositional design processes on two-dimensional surfaces with mono-modal thermal properties [3]. It has gained more foundational knowledge of wood thermal behavior [4, 5]. To advance this design process, we tackle this inverse wood design challenge by reversing the conventional structure-property-behavior pipeline, i.e., thermal-driven wood design, which translates wood thermal behavior to its structure. The motivation is to map a targeted two-dimensional steady-state thermal response directly back to its structural representation, generating a high-fidelity wood design.

Recently, generative models, such as diffusion models [6, 7] and flow matching models [8–10], have achieved unprecedented success in computer vision, revolutionizing image synthesis [6, 11]. Despite these breakthroughs, their translation into the domain of inverse material design (e.g., wood design) remains critically underexplored. The mapping from a thermal response to a physical wood structure is fundamentally an ill-posed, one-to-many problem, as multiple complex microscopic structures can exhibit identical macroscopic thermal footprints. While standard generative models excel at producing visually realistic images, they typically rely on latent-space compression [11–14] that inevitably blurs fine-grained physical boundaries. Consequently, a robust generative pipeline tailored to the pixel-perfect constraints of wood design is still lacking.

To bridge this gap, we frame thermal-driven wood design as a conditional generative task using flow matching. To avoid the lossy nature of latent autoencoders, we operate entirely in pixel space, leveraging Pixel Diffusion Transformers (PixelDiT) [15]. We extract robust spatial semantic conditions using a frozen DINOv3 [16] encoder to guide the generative process, establishing a deterministic probability flow that maps the target thermal layout to the required micro-structural wood patterns (Fig. 1).

2 Related Work

Pixel Space Diffusion Models. Latent diffusion [11] has been a standard solution to image generation due to its computational efficiency using a compression-expansion autoencoder [17],

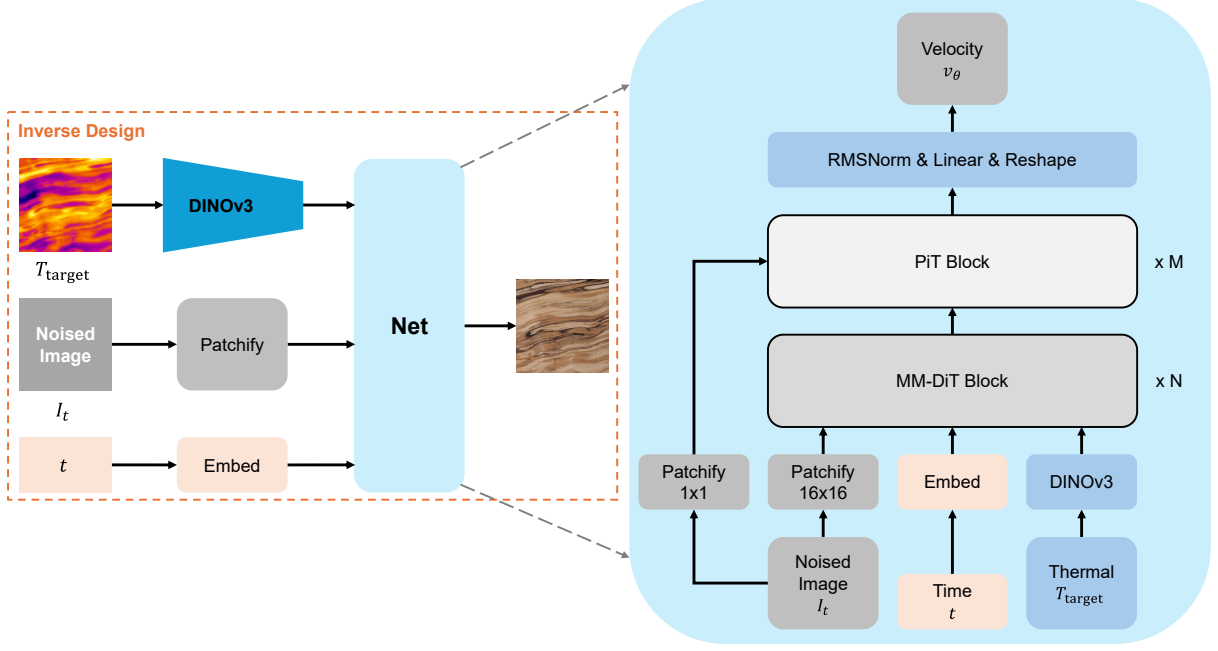


Figure 1: Overview of Flow Matching for Thermal-Driven Wood Design. We leverage DINOv3 [16] to extract semantic features from thermal responses as conditions and the PixelDiT [15] backbone to guide wood design generation.

which inevitably blurs fine-grained details. However, the original diffusion models operate in pixel space [6, 18–20] based on U-Net [21]. With the development of data and models, researchers have shifted to transformer-based diffusion models [15, 22, 23] using vision transformers [24] for better scalability.

Representation Autoencoders for Image Generation. Unlike [25, 26] which utilize representation autoencoders (e.g., DINO-based [16, 27] autoencoders) for image generation, we leverage DINO features and pixel transformers [15] to guide wood design generation.

3 Flow Matching

Let $\mathbf{x}$ represent a vector. We aim to construct a probability path $p_t(\mathbf{x})$ for $t \in [0, 1]$ from Gaussian noise $p_0(\mathbf{x}) = \mathcal{N}(\mathbf{x}; 0, \mathbf{I})$ to an empirical data distribution $p_1(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$. In the flow matching framework [8, 9], the path $\mathbf{x}_t$ between a noise sample $\mathbf{x}_0$ and a data sample $\mathbf{x}_1$ is defined as a straight line $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$, and differentiating the path $\mathbf{x}_t$ with respect to time t gives its target vector field $\mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0, \mathbf{x}_1) = \mathbf{x}_1 - \mathbf{x}_0$. The flow matching objective is to minimize the discrepancy between the target vector field $\mathbf{u}_t$ and the estimated vector field $\mathbf{v}_\theta$, $\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} [\|\mathbf{v}_\theta(\mathbf{x}_t, t \mid \mathbf{c}) - (\mathbf{x}_1 - \mathbf{x}_0)\|_2^2]$, where $\mathbf{v}_\theta(\mathbf{x}_t, t \mid \mathbf{c})$ is parameterized using a neural network given a task-specific conditioning variable $\mathbf{c}$.

4 Flow Matching for Thermal-Driven Wood Design

To perform thermal-driven wood design, we set the evolving state to represent the noised wood RGB image ($\mathbf{x}_t = \mathbf{I}_t \in \mathbb{R}^{3 \times H \times W}$), and the vector field $\mathbf{v}_\theta$ is conditionally guided by a target

thermal response ($\mathbf{c} = \mathbf{T}^\dagger \in \mathbb{R}^{1 \times H \times W}$)

$$\frac{d\mathbf{I}_t}{dt} = \mathbf{v}_\theta(\mathbf{I}_t, t \mid \mathbf{T}^\dagger), \quad (1)$$

with an initial condition of $\mathbf{I}_0 \sim \mathcal{N}(0, \mathbf{I})$, and its solution is given by

$$\mathbf{I}_1 = \mathbf{I}_0 + \int_0^1 \mathbf{v}_\theta(\mathbf{I}_t, t \mid \mathbf{T}^\dagger) dt. \quad (2)$$

Thermal-driven wood design presents a strict domain asymmetry. The conditioning variable $\mathbf{c}$ is a single-channel thermal response $\mathbf{T}^\dagger$, while the target is a three-channel RGB image $\mathbf{I}$. As illustrated in Fig. 1, the target thermal response $\mathbf{T}^\dagger$ is passed through a frozen DINOv3 encoder [16] to extract robust spatial tokens. The DINOv3 encoder is pre-trained on three-channel images. We standardize the thermal response $\mathbf{T}^\dagger$ using dataset statistics (μ, σ) and duplicate the spatial dimensions across the channel axis

$$\mathbf{T}_{\text{cond}} = \left[\frac{\mathbf{T}^\dagger - \mu}{\sigma} \right]_{\times 3} \in \mathbb{R}^{3 \times H \times W}. \quad (3)$$

This aligned condition is passed through a frozen DINOv3 encoder with a patch size of $p = 16$ to extract a sequence of high-fidelity wood thermal layout tokens

$$\mathbf{z}_{\text{cond}} = \text{DINOv3}(\mathbf{T}_{\text{cond}}) \in \mathbb{R}^{L \times D_{\text{dino}}}, \quad (4)$$

where $L = (H/p)(W/p)$ represents the sequence length and D_{dino} denotes the DINOv3 encoder’s intrinsic hidden dimension. We use a DINO-Guided Pixel Diffusion Transformer based on PixelDiT [15] to accommodate this asymmetry. To optimize the vector field $\mathbf{v}_\theta$ with stable numerical integration, the target RGB image $\mathbf{I}$ is linearly scaled to a symmetric $[-1, 1]$ space during the training phase.

Macro-Physics Patch-Level Transformer. Following [15], the patch-level pathway computes the macroscopic trajectory. The evolving noisy three-channel state $\mathbf{I}_t \in \mathbb{R}^{3 \times H \times W}$ is patchified into $p \times p$ nonoverlapping patches and projected into a sequence, $\mathbf{s}_0 \in \mathbb{R}^{L \times D}$, where D is the hidden dimension. The scalar integration time t is embedded via a Multi-Layer Perceptron (MLP) into $\mathbf{t}_{\text{emb}} \in \mathbb{R}^{1 \times D}$. Within the patch-level blocks, $\mathbf{t}_{\text{emb}}$ modulates the tokens via adaptive layer normalization (AdaLN). We replace the text tokens with our thermal tokens. To anchor the generative trajectory to the thermal target, the network utilizes a multi-modal diffusion transformer-based (MM-DiT) backbone [13] with 2D Rotary Position Embeddings (RoPE) to establish spatial coordinate awareness over the noise image, yielding a final sequence of macro-semantic instructions, $\mathbf{s}_{\text{cond}} = \mathbf{s}_N + \mathbf{t}_{\text{emb}} \in \mathbb{R}^{L \times D}$.

Micro-Physics Pixel-Level Transformer. Following [15], the micro-physics pathway refines the individual pixels using a pixel transformer-based (PiT) backbone based on the macro-instructions $\mathbf{s}_{\text{cond}}$. The high-resolution state $\mathbf{I}_t$ is projected into a compact latent pixel dimension $D_{\text{pix}} = 16$, grouping the batch and sequence dimensions to process localized pixel patches independently, $\mathbf{z}_0 \in \mathbb{R}^{L \times p^2 \times D_{\text{pix}}}$. An MLP directly maps the semantic token of the i -th patch $\mathbf{s}_{\text{cond}}^{(i)}$ into unique scale (γ) and shift (β) parameters for every individual pixel within that spatial block. These parameters modulate the pixels via pixel-wise AdaLN

$$\mathbf{z}_{\text{mod}} = \gamma(\mathbf{s}_{\text{cond}}) \odot \text{RMSNorm}(\mathbf{z}_0) + \beta(\mathbf{s}_{\text{cond}}) \in \mathbb{R}^{L \times p^2 \times D_{\text{pix}}}. \quad (5)$$

To ensure the local gradients maintain global consistency, the p^2 pixels within each patch are linearly compacted into a single token, processed by global self-attention over the L length sequence, and expanded back to the high-resolution grid. A final 1×1 convolutional layer projects the D_{pix} channels strictly back to the three channels.

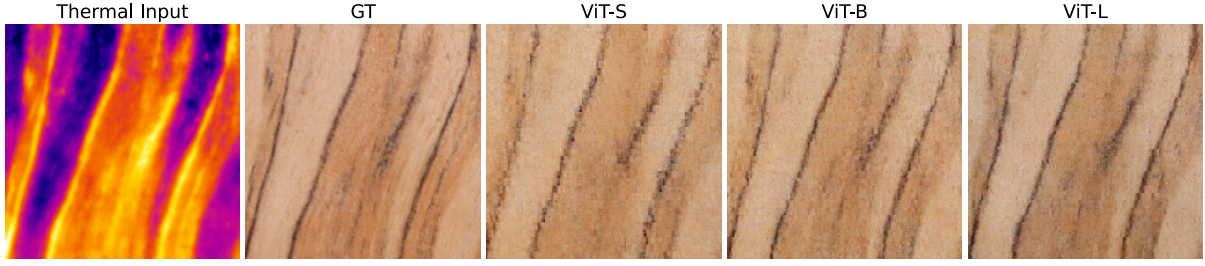


Figure 2: The Results of Flow Matching for Thermal-Driven Wood Design Using DINOv3 ViT-S, ViT-B, and ViT-L. A key observation is that as we increase the model size, we improve the quality of generation.

DINO-Guided Pixel Diffusion Transformer. Our DINO-Guided Pixel Diffusion Transformer consists of N MM-DiT blocks and M PiT blocks. The patch-level macro-network establishes the bulk flow trajectory by integrating this wood thermal layout via joint multi-head cross-attention in the MM-DiT. These macro-instructions are then passed to the pixel-level network, which utilizes pixel-wise AdaLN to individually modulate the evolving high-frequency pixels in the PiT, ensuring that the generated physical boundaries strictly obey localized physics without scaling attention quadratically to predict the vector field $\mathbf{v}_\theta(\mathbf{I}_t, t \mid \mathbf{T}^\dagger)$.

5 Experiments

We use DINOv3 encoders (ViT-S, ViT-B, and ViT-L) to guide the wood generation (Fig. 2). A key observation is that as we increase the model size, we improve the quality of generation.

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